03. Exploratory Data Analysis

Use EDA to understand language mix, text type and filtering workload.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

EDA and diagnostics in the notebook

- Loads evaluation_data.xlsx and separates rows into article and tweet subsets based on Text_ID.
- Inspects sample article and tweet records.
- Cleans both text types and extracts candidate brands before theme classification.
- Counts predicted Mobile_Tech classes separately for articles and tweets.
- Detects language among the mobile-tech subset and inspects language counts.
- Builds per-brand sentiment output rows and exports a 557-row sentiment-output.csv.

Observed evaluation-sample counts

Text type	Mobile-tech = 0	Mobile-tech = 1	Total
Tweets	148	62	210
Articles	142	71	213
Total	290	133	423

Language mix among predicted mobile-tech subset

Type	English	Hindi	Hinglish	Total
Articles	45	24	2	71
Tweets	32	18	12	62

What these counts mean

The notebook's predicted mobile-tech subset is 71 articles and 62 tweets. The article and tweet totals in the class count output are 213 and 210 respectively, while the earlier dataframe shape shown is 222 articles and 210 tweets; the notebook has some inconsistent counts/outputs due to filtered rows or saved output state. Treat these counts as notebook snapshots, not canonical dataset statistics.

EDA improvements

- Verify unique Text_ID counts at each stage and reconcile 222 versus 213 article rows.
- Inspect duplicate texts, empty strings, very long articles, malformed encodings and language-detection confidence.
- Report class balance on labelled train/validation/test sets, not just predicted evaluation sample.
- Measure brand frequency, multi-brand prevalence, brand alias ambiguity and sentiment distribution.
- Inspect translation quality separately for Hindi and Hinglish.